James Pritts

https://prittjam.github.io/

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Academic Appointments

Christian-Albrechts-Universität zu Kiel — Kiel, Germany — 2024 – Now
KiTE Research Fellow

Ukrainian Catholic University — Lviv, Ukraine — 2023 – 2024
Assistant Professor

Czech Inst. of Informatics, Robotics & Cybernetics — Prague, Czechia — 2022 – 2023
Research Fellow

Chalmers University of Technology — Gothenburg, Sweden — 2021
Post-Doctoral Research Scientist

Facebook Reality Labs, AR/VR — Pittsburgh, PA — 2019 – 2021
Post-Doctoral Research Scientist

Education

Czech Technical University — Prague, Czechia — 2020
Ph.D., Computer Science
Thesis: "Methods for the Rectification of Imaged Coplanar Repeated Patterns"

Czech Technical University — Prague, Czechia — 2013
M.Sc., Computer Science

The University of North Texas — Denton, TX — 2002
B.Sc., Mathematics

Publications

J. Pritts, T. Sittart, H. Sauer, S. Janßen, F. Seegräber, D. Nakath, and K. Köser. BlobBoards: Robust Markers for Accurate Pose. In *BMVC*, 2026

J. Pritts, F. Seegräber, and K. Köser. RANSAC Scoring Done Right. *arXiv preprint arXiv:2606.27385*, 2026

E. Dexheimer, P. Peluse, J. Chen, **J. Pritts**, and M. Kaess. Information-theoretic online multi-camera extrinsic calibration. *IEEE Robotics and Automation Letters*, 7(2):4757–4764, 2022

Y. Lochman, K. Liepieshov, J. Chen, M. Perdoch, C. Zach, and **J. Pritts**. Babelcalib: a universal approach to calibrating central cameras. In *ICCV*, 2021

Y. Lochman, O. Dobosevych, R. Hryniv, and **J. Pritts**. Minimal Solvers for Single-View Lens-Distorted Camera Auto-Calibration. In *WACV*, 2021

J. Pritts, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal solvers for rectifying from radially-distorted conjugate translations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):3931–3948, 2021

J. Pritts, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal Solvers for Rectifying from Radially-Distorted Scales and Change of Scales. *International Journal of Computer Vision*, 128(4):950–968, 2020

J. Pritts, Z. Kukelova, V. Larsson, and O. Chum. Rectification from Radially-Distorted Scales. In *ACCV*, 2018

J. Pritts, Z. Kukelova, V. Larsson, and O. Chum. Radially-Distorted Conjugate Translations. In *CVPR*, 2018

J. Pritts, D. Rozumnyi, M. P. Kumar, and O. Chum. Coplanar Repeats by Energy Minimization. In *BMVC*, 2016

J. Pritts, O. Chum, and J. Matas. Detection, Rectification and Segmentation of Coplanar Repeated Patterns. In *CVPR*, 2014

J. Pritts, O. Chum, and J. Matas. Approximate Models for Fast and Accurate Epipolar Geometry Estimation. In *IVCNZ*, 2013

Recognition and Awards

Asian Conference on Computer Vision (ACCV) Saburo Tsuji Best Paper Award — 2018
for "Rectification from Radially-Distorted Scales"

Image and Vision Computing New Zealand (IVCVNZ) Best Paper Award — 2013
for "Approximate Models for Fast and Accurate Epipolar Geometry Estimation"

Professional Association Memberships

ELLIS	Scholar	2024 – Now
Kiel Marine Science	Associate	2025 – Now

Funding

Principal	Kiel Marine Science (KMS) "Enhancing Underwater Robotics for Visual Mapping in Turbid Environments" Awarded: €10,000	2025
Principal	Kiel Training for Excellence (KiTE) co-funded by MSCA "Live and Robust Visual Localization and Mapping in Coastal Waters" Awarded: $225,000	2024 – 2027
Contributing	Facebook Sponsored Research Agreement "In-the-field Extrinsic Calibration of Multi-camera Systems" Awarded: $150,000	2020 – 2021
Principal	Facebook Sponsored Research Agreement "Calibration of Head-Mounted Multi-Camera Capture Systems" Awarded: $25,000	2020

Professional Service Activities

Reviewer	CVPR, ICCV, ECCV, 3DV, WACV, ACCV, ICRA, IJCV, PAMI	
Conference Organizer	Poster Chair	3DV 2022
Tutorial Organizer	"Affine Correspondences and Their Applications"	3DV 2022
Tutorial Organizer	"Affine Correspondences and Their Applications"	CVPR 2022

Teaching

3D Computer Vision — 2025 – 2026
Substitute Instructor, M.Sc. level, Kiel University

Image Retrieval — 2017 – 2018
Instructor, M.Sc. level, Ukrainian Catholic University

Pattern Recognition and Machine Learning — 2013 – 2016
Teaching Assistant, B.Sc. level, Czech Technical University in Prague

Advising

M.Sc. Students

Ties Harder	Kiel University Thesis: “Boundary Movement Aware Differentiable Rendering for Visual Calibration Tasks”	in progress
Philipp Böll	Kiel University Thesis: “Underwater Light Rig Auto-Calibration from Asynchronous Cameras”	in progress
Hendrik Sauer	Kiel University Thesis: “Affine-Equivariant Neural Networks for the Feature Matching Problem”	2026
Till Sittart	Kiel University Thesis: “Affine-Equivariant Blob Targets for Robust Absolute Pose in Robotics”	2026
Liam Boddin	Kiel University Thesis: “Underwater Calibration of Near-point Anisotropic Lights”	2025
Igor Babin	Ukrainian Catholic University Thesis: “Image Inpainting in Latent Space”	2024
Dmytro Nadobko	Ukrainian Catholic University Thesis: “Supervised Learning of Correspondence Volumes for Coplanar Repetitive Patterns”	2023
Andrii Stadnik	Ukrainian Catholic University Thesis: “Corner Localization and Camera Calibration from Imaged Lattices”	2023
Yaroslava Lochman	Ukrainian Catholic University Thesis: “Minimal Solvers for Single-View Auto-Calibration” *moved on as Ph.D. student at Chalmers University of Technology*	2018 – 2020

B.Sc. Students

Yaroslav Romanus	Ukrainian Catholic University Thesis: “Applying Motion in Latent Space”	2024
Ostap Viniavskyi	Ukrainian Catholic University Thesis: “Learning Discriminative Context-Aware Keypoints Representations for Resolving Ambiguous Matches” *moved on as Researcher at The ML Lab at Ukrainian Catholic University*	2019 – 2021
Kostiantyn Liepieshov	Ukrainian Catholic University Thesis: “Manhattan-Frame Detection in Lens-Distorted Images” *moved on as M.Sc. student at Ukrainian Catholic University*	2019 – 2021

Invited Talks and Lectures

3DV Tutorial Talk — September 12, 2022
"Just One Image is All It Takes..." — Prague, Czechia

CVPR Tutorial Talk — June 19, 2022
"Just One Image is All It Takes..." — New Orleans, LA

Facebook Reality Labs — June 14, 2018
"Minimal Solvers and Multi-Model Estimators for Image Undistortion and Scene-Plane Rectification" — Pittsburgh, PA

The Aspen Institute's 2018 Young Leader's Program — March 17, 2018
"Opportunities and Risks of Artificial Intelligence" — Tále, Slovakia

The Eastern European Computer Vision Conference — July 14, 2017
"Detection, Rectification, and Segmentation of Coplanar Repeated Patterns" — Odesa, Ukraine

The 34th Pattern Recognition and Computer Vision Colloquium — April 3, 2014
"Detection, Rectification, and Segmentation of Coplanar Repeated Patterns" — Prague, Czechia